

ALEKSANDROV-GAJ, Russian Federation

WMO: 343910

Lat: 50.150N

Lon: 48.550E

Elev: 81

StdP: 14.65

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-9.0	-20.3	1.8	-13.8	-14.7	2.5	-8.2	23.2	25.7	21.2	28.0	5.1	70	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	98.4	68.8	95.0	68.0	91.5	67.1	70.9	91.4	69.7	89.8	68.4	87.1	7.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	93.2	75.7	63.5	87.8	75.3	61.9	83.0	75.2	34.7	91.3	33.7	89.4	32.7	87.1	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	19.8	17.5	15.5	DB	-19.8	102.1	6.5	3.0	-24.5	104.3	-28.3	106.1	-31.9	107.7	-36.6	109.9
(5)				WB	-20.1	73.2	6.4	1.8	-24.7	74.5	-28.5	75.5	-32.1	76.5	-36.8	77.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	15.2	16.0	29.4	49.2	61.8	71.9	76.1	73.2	60.6	46.4	31.4	20.0	
	DBStd	24.03	12.22	13.07	10.53	9.06	8.60	7.89	6.72	7.81	8.43	8.23	10.24	12.12	
	HDD50	4472	1078	951	639	122	11	0	0	0	12	171	559	929	
	HDD65	7866	1543	1371	1103	477	164	24	3	19	184	576	1008	1394	
	CDD50	3056	0	0	1	98	378	658	810	721	328	61	1	0	
	CDD65	974	0	0	0	3	66	232	348	274	50	1	0	0	
	CDH74	12127	0	0	0	58	842	2833	4277	3407	695	15	0	0	
	CDH80	5836	0	0	0	7	306	1380	2193	1711	237	2	0	0	
Wind	WSAvg	7.3	7.7	8.5	8.3	7.8	7.3	6.8	6.4	6.4	6.6	7.1	7.6	7.8	
Precipitation	PrecAvg	12.4	1.2	1.0	0.7	0.8	0.9	1.3	1.1	1.1	0.9	1.0	1.3	1.2	
	PrecMax	17.2	2.5	2.9	1.9	3.0	2.8	5.2	4.5	3.5	2.5	2.1	3.0	2.6	
	PrecMin	6.9	0.3	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.1	0.2	0.2	0.4	
	PrecStd	3.1	0.6	0.7	0.5	0.7	0.7	1.0	1.0	0.8	0.6	0.5	0.9	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	39.6	60.9	80.2	91.8	101.3	102.5	101.8	91.2	75.7	56.4	44.0	
		MCWB	39.6	36.9	49.1	60.1	64.6	68.9	69.8	69.9	63.8	56.4	51.0	42.4	
	2%	DB	36.4	36.0	52.7	74.6	87.3	96.7	98.5	97.4	86.1	70.5	51.2	39.4	
		MCWB	35.5	34.3	45.4	57.0	64.3	67.6	69.2	68.8	62.0	55.0	47.9	37.8	
	5%	DB	33.7	34.6	47.0	69.7	83.3	92.4	94.9	93.9	82.0	64.8	47.5	36.5	
		MCWB	32.9	33.3	41.8	55.6	62.7	67.1	68.7	67.5	60.7	52.7	44.7	35.4	
	10%	DB	32.1	33.2	42.2	64.8	78.6	88.2	91.2	89.6	77.3	60.0	44.3	34.1	
		MCWB	31.5	32.3	38.5	52.3	60.7	66.0	67.6	66.2	59.3	51.2	42.4	33.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.8	37.5	50.2	62.4	67.6	71.8	72.9	72.0	67.5	61.8	52.2	42.9
			MCDB	40.3	39.1	57.7	76.0	85.4	93.5	92.6	93.0	84.6	68.5	54.6	43.8
2%		WB	35.7	34.9	45.8	58.8	65.9	69.9	71.1	70.3	64.4	57.0	48.4	38.3	
		MCDB	36.3	35.8	51.8	71.1	83.9	89.8	91.3	91.9	80.8	65.5	50.9	39.0	
5%		WB	33.1	33.6	42.0	56.3	63.6	68.3	70.1	68.9	62.4	54.4	45.1	35.5	
		MCDB	33.5	34.5	46.3	67.3	79.6	87.6	89.7	89.4	78.3	62.4	46.9	36.4	
10%		WB	31.5	32.4	38.7	53.6	61.7	67.0	68.9	67.5	60.3	51.9	42.5	33.3	
		MCDB	32.1	33.3	41.9	63.4	76.5	85.0	87.2	86.1	73.4	58.9	44.3	33.9	
Mean Daily Temperature Range	5% DB	MDBR	11.7	13.4	14.2	20.3	22.0	22.5	22.4	23.8	23.6	18.2	11.8	11.0	
		MCDBR	7.5	9.4	20.4	26.8	27.7	27.7	26.7	28.2	30.3	26.6	15.1	8.7	
		MCWBR	7.5	8.3	13.4	13.0	11.1	9.6	9.1	10.7	13.4	14.2	11.7	8.0	
	5% WB	MCDBR	7.6	8.7	19.3	24.5	25.0	24.7	23.7	25.4	26.5	22.1	13.4	8.4	
		MCWBR	7.7	8.0	13.2	12.7	11.3	9.7	8.9	10.2	12.7	13.5	11.2	8.1	
Clear-Sky Solar Irradiance	taub	0.322	0.346	0.387	0.448	0.423	0.404	0.413	0.430	0.379	0.358	0.337	0.319		
	taud	2.074	2.018	2.083	2.043	2.201	2.316	2.291	2.223	2.332	2.324	2.265	2.135		
	Ebn at Noon	212	239	254	251	265	271	266	254	257	239	211	194		
	Edh at Noon	27	37	42	49	43	39	39	40	32	27	23	22		
All-Sky Solar Radiation	RadAvg	345	711	1082	1465	1928	2082	2019	1743	1236	709	328	226		
	RadStd	81	122	156	112	139	174	134	118	121	94	47	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	+3.35	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+2.21	N/A	N/A	N/A	N/A	+299	+196

Nomenclature: See separate page